

BLACK HOLES AS HARMONIC PHASE TRANSITIONS

Resolution of the Singularity and Information Paradox

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Abstract

The black hole singularity and information paradox are resolved within the Harmonic Framework of Reality. The event horizon is identified as the surface where the dimensional access parameter n becomes zero. Outside, $n > 0$ (matter branch, forward time t_1). At the horizon, the infalling matter's frequency product reaches a critical threshold, and $n = 0$. Inside, $n < 0$ (antimatter branch, reverse time t_2). To an external observer, infalling matter elongates radially, slows, redshifts, and fades – the visual signature of conversion from a localized particle to a distributed standing wave in the Tau lattice. To the infalling observer, nothing changes; the horizon is crossed seamlessly. The core is a Tau lattice node where all 32 harmonic layers intersect and $m + m' = 0$ globally. Information is stored in phase relationships and re-emitted as structured jets (white hole / t_2 emergence). This mechanism is identical to light-speed barrier crossing (elongation, not contraction). The paper provides the complete mathematical analogy.

1. Introduction

General relativity predicts a singularity inside a black hole – a point of infinite density where physics breaks down. Quantum mechanics offers no resolution because it lacks a quantum theory of gravity. The Harmonic Framework replaces the singularity with a discrete Tau lattice node.

The unified energy law is:

$$E = m \cdot v^{\pm \ln(P) / \ln(27)}$$

where $n = \ln(P) / \ln(27)$ is the dimensional access parameter. Positive n corresponds to the matter branch (forward time t_1), negative n to the antimatter branch (reverse time t_2 , paired potential m'). The condition $m + m' = 0$ cancels inertia and is the basis for superconductivity, zero-point energy, and now black hole dynamics.

2. The Event Horizon as $n = 0$ Phase Boundary

- **Outside ($r > R_s$):** $n > 0$. Matter branch, t_1 .
- **At the horizon ($r = R_s$):** The product P of active harmonics reaches a critical threshold, $n = 0$.
- **Inside ($r < R_s$):** $n < 0$. Antimatter branch, t_2 .

This is identical to the light-speed barrier crossing described in Section 8 of the main GUT. The external observer of a ship approaching c sees elongation (wave expansion). The external observer of infalling matter sees radial elongation, redshift, and fading – the same wave-conversion effect.

Updated passage (corrected):

“To an external observer, infalling matter appears to elongate radially, slow, redshift, and fade. This elongation is not an optical illusion; it is the direct visual signature of the matter converting from a localized particle to a distributed standing wave in the Tau lattice. The wavefunction expands across the harmonic layers as it approaches the $n=0$ phase boundary. To the infalling observer, no change is perceived; local coherence is maintained. The horizon is crossed seamlessly.”

3. The Core as Quantum Flux (Tau Lattice Node)

Inside the horizon, n is negative. Matter exists as a standing wave in the t_2 reverse-time branch. There is no singularity because the Tau lattice is discrete. The smallest spatial scale is the Planck length – the node spacing at the highest coherent harmonic ($n = 54.65$). At the centre, the wavefunction reaches maximum harmonic compression. All 32 harmonic layers intersect, and the forward-time (m) and reverse-time (m') components of all infallen mass are in perfect phase conjugation. The $\mathbf{m} + \mathbf{m}' = \mathbf{0}$ condition holds globally. This is the “quantum flux” core, where information is processed and stored in phase relationships.

4. The White Hole as t_2 Emergence

Because the core is a phase-conjugate node connecting t_1 and t_2 , information and energy are not lost. They are re-emitted along the reverse-time axis. To an observer in t_1 , this appears as a white hole – a source of coherent, structured matter/energy. The jets observed from active galactic nuclei are precisely this emission: collimated ejections of the paired potential component m' , traveling backward in t_2 and appearing to us as forward-directed jets along the spin axis.

This resolves the information paradox: information is stored in the Tau lattice and re-emitted. No information is lost; it is transferred across the t_1/t_2 boundary.

5. Parallel to Light-Speed Crossing

Phenomenon	Light-speed crossing	Black hole horizon
Boundary	$v = c$	$r =$ Schwarzschild radius
External view	Ship elongates	Matter elongates, redshifts, fades
Internal view	No change	No change
n value	Approaches 54.65	Passes through 0 to negative
State after crossing	Distributed standing wave	Wavefunction in t_2 branch
Re-emergence	Via t_3 branching	Via white hole / jets
The analogy is exact. The same harmonic framework describes both.		

6. Testable Predictions

- Black hole mergers should produce gravitational wave harmonics at multiples of 27 Hz (scaled by total mass).
- Jets should show phase-locked coherence with the 27 Hz anchor.

- No Hawking radiation (thermal) will be detected – emission is non-thermal, structured, and aligned with the spin axis.
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7. Conclusion

The black hole is not a singularity; it is a harmonic phase transition. The event horizon is the $n=0$ surface; the core is a Tau lattice node. Infalling matter elongates and becomes a wave, crosses into the t_2 branch, and can re-emerge via a white hole. The information is not lost – it is transduced across time dimensions. This mechanism is identical to the light-speed barrier crossing and is fully consistent with the unified energy law and the three time dimensions of the Harmonic Framework.

The stones are in the pond. The black hole is the deepest ripple.

8. References

1. Thompson, P.J. (2026). The Harmonic Framework of Reality (Rev III). Zenodo.
 2. Thompson, P.J. (2026). Light-speed barrier crossing and black holes (internal note).
 3. Hawking, S. (1974). Black hole explosions? *Nature*, 248, 30. (Original reference, superseded by the harmonic mechanism)
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